

BACK-UP COMPUTATION

(Variation Order No. 1)

Construction of Covered Court

B.3(1) - Permits and Clearances

Orig Qty. = **1.00 L.S.**
V.O. 1 Qty. = **1.00 L.S.**

B.5(1) - Project Billboard/Signboard

Orig Qty. = **2.00 each**
V.O. 1 Qty. = **2.00 each**

B.7(1) - Occupational Safety & Health Program

Orig Qty. = **1.00 L.S.**
V.O. 1 Qty. = **1.00 L.S.**

B.8(1) - Traffic Management

Orig Qty. = **7.00 mo.**
V.O. 1 Qty. = **7.00 mo.**

B.9(1) - Mobilization/Demobilization

Orig Qty. = **1.00 L.S.**
V.O. 1 Qty. = **1.00 L.S.**

B.24(1) - General scaffolding and shoring and netting

Orig Qty. = **1.00 L.S.**
V.O. 1 Qty. = **1.00 L.S.**

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****801(1) - Removal of Structure and Obstruction**

Orig Qty. = **1.00 L.S.**
 V.O. 1 Qty. = **1.00 L.S.**

803(1)a - Structure excavation (Common Soil)

Orig Qty. = **242.70 cu.m.**
 V.O. 1 Qty. = **244.05 cu.m.**

C1F1 =	2.60	X	2.60	X	1.90	X	12.00
=	154.13	m3					
FTB-1 =	0.50	x	0.90	x	22.00	x	6.00
=	59.40	m3					
FTB-2 =	0.50	x	0.90	x	7.00	x	10.00
=	28.00	m3					
RAMP =	9.00	x	0.70	x	0.40	x	1.00
=	2.52	m3					
V TOTAL =	244.05	m3					

804(1)a - Embankment from Roadway/Structure Excavation (Common Soil)

Orig Qty. = **135.04 cu.m.**
 V.O. 1 Qty. = **140.61 cu.m.**

C1F1 =	2.60	x	2.60	x	1.40	x	12.00
=	113.57	m3					
C1 =	0.50	x	0.50	x	1.40	x	12.00 (deduct)
=	4.20	m3					
FTB-1 =	0.50	x	0.30	x	22.00	x	6.00
=	19.80	m3					
FTB-2 =	0.50	x	0.30	x	7.00	x	10.00
=	10.50	m3					
Ramp =	0.15	x	0.35	x	9.00	x	2.00
=	0.95	m3					
VTOTAL =	140.61	m3					

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****804(2)a - Embankment from Borrow(Common soil)**Orig Qty. = **308.00 cu.m.**V.O. 1 Qty. = **276.78 cu.m.**

V(Slab on Fill) =	22.00	x	35.00	x	0.30
=	231.00	m2			
Sides =	0.63	x	35.00	x	2.00
=	43.75	m2			
Ramp =	6.00	x	1.50	x	0.15
=	1.35	m2			
=	1.50	x	1.50	x	0.30
=	0.68				
V Total =	276.78 m.	m2			

804(7) - GravelfillOrig Qty. = **80.30 cu.m.**V.O. 1 Qty. = **94.32 cu.m.**

C1F1 =	2.60	X	2.60	x	0.10	x	12.00
=	8.11	m3					
FTB-1 =	0.40	x	22.00	x	0.10	x	6.00
=	5.28	m3					
FTB-2 =	0.40	x	7.00	x	0.10	x	10.00
=	2.80	m3					
V(Slab on Fill) =	22.00	x	35.00	x	0.10	x	1.00
=	77.00	m3					
Ramp =	7.50	x	1.50	x	0.10	x	1.00
=	1.13	m3					
VTOTAL =	94.32	m3					

900(1)a - Structural Concrete (Class A, 3000psi @ 7 days)Orig Qty. = **92.56 cu.m.**V.O. 1 Qty. = **108.13 cu.m.**

C1F1 =	2.50	x	2.50	x	0.40	x	12.00
=	30.00	m3					
V(Slab on Fill) =	22.00	x	35.00	x	0.10	x	1.00
=	77.00	m3					
Ramp =	1.50	x	7.50	x	0.10	x	1.00
=	1.13	m3					
VTOTAL =	108.13						

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****900(1)b - Structural Concrete (Class A, 3000psi @ 14 days)**Orig Qty. = **62.56 cu.m.**V.O. 1 Qty. = **61.10 cu.m.**

FTB-1	=	0.40	X		0.50	X	22.00	X	6.00
	=	26.40	m3						
FTB-2	=	0.40	x		0.50	x	7.00	x	10.00
	=	14	m3						
C1	=	0.50	x		0.50	x	6.90	x	12.00
	=	20.7	m3						
VTOTAL	=	61.10	m3						

902(1)a1 - Reinforcing Steel, (Deformed, Grade 40)Orig Qty. = **18,608.77 kgs.**V.O. 1 Qty. = **18,980.73 kgs.**

C1	=	216.00	-	25 mm Ø x 6.00 m	=	4993.95 kgs.
	=	539.00	-	10 mm Ø x 6.00 m	=	1993.88 kgs.
						6987.84 kgs.
C1F1	=	168.00	-	20 mm Ø x 6.00 m	=	2485.88 kgs.
						2485.88 kgs.
FTB-1	=	230.00	-	20 mm Ø x 6.00 m	=	3403.29 kgs.
	=	496.00	-	10 mm Ø x 6.00 m	=	1834.82 kgs.
						5238.10 kgs.
FTB-2	=	69.00	-	20 mm Ø x 6.00 m	=	1020.99 kgs.
	=	116.00	-	10 mm Ø x 6.00 m	=	429.11 kgs.
						1450.10 kgs.
WF-1	=	6.00	-	12 mm Ø x 6.00 m	=	88.78 kgs.
	=	3.00	-	10 mm Ø x 6.00 m	=	11.10 kgs.
						99.88 kgs.
Slab on Fill	=	735.00	-	10 mm Ø x 6.00 m	=	2718.93 kgs.
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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****1046(2)a1 - CHB Non-Load Bearing (Including Reinforcing Steel, 100mm)**Orig Qty. = **52.00 sq.m.**V.O. 1 Qty. = **20.58 sq.m.**

Ramp	=	0.35	x	8.50	x	1.00
	=	17.70	m2			
	=	2.88	x	1.00	x	1.00
	=	2.88	sq.m.			
	=	20.58	sq.m.			

1046(2)a2 - CHB Non-Load Bearing (Including Reinforcing Steel, 150mm)Orig Qty. = **54.60 sq.m.**V.O. 1 Qty. = **91.20 sq.m.**

Court	=	114.00	x	0.80	x	1.00
		91.20	m2			
A(Total)	=	91.20	m2			

1027(1) - Cement Plaster FinishOrig Qty. = **150.90 sq.m.**V.O. 1 Qty. = **184.50 sq.m.**

C1	=	2.00	x	5.00	x	12.00
	=	120.00	m2			
Embankment Wall	=	114.00	x	0.50	x	1.00
	=	57.00	m2			
Ramp	=	15.00	x	0.50	x	1.00
	=	7.50	m2			
	=	184.50	m2			

1003(17) - Carpentry and Joinery Works

(new scope of Work)

Orig Qty. = **1.00 L.S.**V.O. 1 Qty. = **1.00 L.S.**

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****1032(1)c - Painting Works(Steel)**Orig Qty. = **1,342.89 sq.m.**V.O. 1 Qty. = **1,334.75 sq.m.**

A1	=	2.50	x	100.00	Steel Angle, 2-4.0"x4.0"x8mm
	=	249.60	m2		
A2	=	1.88	x	103.00	Steel Angle, 2-3.0"x3.0"x6.35mm
	=	193.25	m2		
A3	=	1.20	x	53.00	Steel Angle, 2-48mmx48mmx4.0mm
	=	63.60	m2		
A4	=	0.96	x	73.00	Steel Angle, 2-38mmx38mmx4.0mm
	=	70.08	m2		
A5	=	3.02	x	220.00	C-purlins- 2"x 6"x6.0m GA 14
	=	665.28	m2		
A6	=	4.82	x	11.00	Fascia- 2" x 12" x 6.0m GA# 14
	=	53.06	m2		
A7	=	1.86	x	4.00	Angle Cleats-75mmx75mmx4.75mm
	=	7.43	m2		
A8	=	0.52	x	12.00	Steel Base Plate - 0.5m x 0.5m x 12mm thick
	=	6.29	m2		
A9	=	0.06	x	370.00	Gusset Plate - 0.1m x 0.250m x 10mm thick
	=	21.09	m2		
A10	=	0.00068	x	18.00	Sagrods,12mm x6.0mm
	=	0.01	m2		
A11	=	0.00121	x	40.00	Tension Rods,16mm x6.0m
	=	0.05	m2		
A11	=	0.71628	x	7.00	38mm G.I pipe
	=	5.01	m2		
Total	=	1334.75	m2		

1014(1)a2 - Prepainted Metal Sheets(above 0.427mm thk., Corrugated, Long Span)Orig Qty. = **995.30 sq.m.**V.O. 1 Qty. = **807.00 sq.m.**

A1	=	26.90	x	30.00
	=	807.00	sq.m	
VTOTAL	=	807.00	sq.m	

1013(2)a - Fabricated Metal Roofing Accessory(Gauge (26(0.551mm), Ridge Rolls/ Hip)Orig Qty. = **37.00 L.M.**V.O. 1 Qty. = **30.00 L.M.**

L = 30.00 L.M

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****1047(8)a - Structural Steel (Trusses)**

Orig Qty. = **17,111.57 kgs.**
 V.O. 1 Qty. = **14,392.88 kgs.**

TRUSS-1

100.00 pcs. - Steel Angle, 2-4.0"x4.0"x8mm = 7536.00 kgs.
 103.00 pcs. - Steel Angle, 2-3.0"x3.0"x6.35mm = 4694.80 kgs.

BEAM-1

21.00 pcs. Steel Angle, 2-48mmx48mmx4.0mm = 379.81 kgs.
 31.00 pcs. Steel Angle, 2-38mmx38mmx4.0mm = 443.87 kgs.

BEAM-2

21.00 pcs. - Steel Angle, 2-48mmx48mmx4.0mm = 379.81 kgs.
 29.00 pcs. - Steel Angle, 2-38mmx38mmx4.0mm = 524.51 kgs.

BEAM-3

11.00 pcs. - Steel Angle, 2-48mmx48mmx4.0mm = 198.95 kgs.
 13.00 pcs. - Steel Angle, 2-38mmx38mmx4.0mm = 235.12 kgs.
 14392.88 kgs.

1047(8)b - Structural Steel (Purlins)

Orig Qty. = **7,454.16 kgs.**
 V.O. 1 Qty. = **6,600.12 kgs.**

220.00 - C-purlins- 2"x 6"x6.0m GA 14 = 6,009.96 kgs.
 11.00 - Fascia- 2" x 12" x 6.0m GA# 14 = 455.93 kgs.
 4.00 pcs. - Angle Cleats-75mmx75mmx4.75mm = 134.24 kgs.
 6,600.12 kgs.

1047(5)d - Metal Structure Accessories (Steel plates)

Orig Qty. = **1,130.40 kgs.**
 V.O. 1 Qty. = **1,008.73 kgs.**

12.00 pcs. - Steel Base Plate - 0.5m x 0.5m x 12mm thick = 282.6 kgs.
 370.00 pcs. - Gusset Plate - 0.1m x 0.250m x 10mm thick = 726.125 kgs.
 1008.725 kgs.

1047(4)a - Metal Structure Accessories (Bolts and Rods)

Orig Qty. = **48.00 each**
 V.O. 1 Qty. = **48.00 each**

48.00 pcs. = 3/4" 16"- Anchor Bolts

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BACK-UP COMPUTATION*(Variation Order No. 1)***Construction of Covered Court****1047(5)b - Metal Structure Accessories (Sagrods)**

Orig Qty. = **95.84 kgs.**
V.O. 1 Qty. = **95.84 kgs.**

18.00 pcs. - Sagrods, 12mm x 6.0mm = 95.84 kgs.

1047(5)c - Metal Structure Accessories (Crossbracing)

Orig Qty. = **378.61 kgs.**
V.O. 1 Qty. = **378.61 kgs.**

40.00 pcs. = Tension Rods, 16mm x 6.0m = 378.61 kgs.

1047(4)b - Metal Structure Accessories (Turnbuckle)

Orig Qty. = **20.00 each**
V.O. 1 Qty. = **16.00 each**

Turnbuckles, 15mm = 16 e.a.

1038(1) - Reflective Insulation

Orig Qty. = -
V.O. 1 Qty. = **807.00 sq.m**

A1 = 26.90 x 30.00
= 807.00 sq.m
VTOTAL = 807.00 sq.m

1100(10) - Conduit, Boxes & Fittings (Conduit Works, Conduit Rough-in)

Orig Qty. = -
V.O. 1 Qty. = **1.00 L.S.**

100(3)a2 - Individual Removal of Trees (Small, 301-500mm dia)

Orig Qty. = -
V.O. 1 Qty. = **5.00 each**

1051(1)a - Railing

Orig Qty. = -
V.O. 1 Qty. = **1.00 L.S.**

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